



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Wednesday, 17 September 2025, 11:08 AM**State** Finished**Completed on** Wednesday, 17 September 2025, 11:10 AM**Time taken** 1 min 2 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int findPair(int arr[], int left, int right, int x, int *a, int *b) {
3      if (left >= right)
4          return 0; // No pair found
5
6      int sum = arr[left] + arr[right];
7
8      if (sum == x) {
9          *a = arr[left];
10         *b = arr[right];
11         return 1;
12     } else if (sum < x) {
13         return findPair(arr, left + 1, right, x, a, b); // Search right half
14     } else {
15         return findPair(arr, left, right - 1, x, a, b); // Search left half
16     }
17 }
18 int main() {
19     int n;
20     scanf("%d", &n);
21     int arr[n];
22     for (int i = 0; i < n; i++)
23         scanf("%d", &arr[i]);
24     int x;
25     scanf("%d", &x);
26     int a, b;
27     if (findPair(arr, 0, n - 1, x, &a, &b)) {
28         printf("%d\n%d\n", a, b);
29     } else {
30         printf("No\n");
31     }
32     return 0;
33 }
34

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			

	Input	Expected	Got	
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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